Ⅱ. 柳彬习题　1. 初等积分法

Ⅱ.1.1. 求特解　(1) $xy\,dx + (x+1)\,dy = 0$, $y(1) = 2$

(2) $(x^2-1)y' + 2xy^2 = 0$　$y(0) = 1$

(3) $xy' + y = y^2$.　$y(1) = \frac{1}{2}$

(4) $y' = \sqrt{4x+2y-1}$　$y(0) = 1$

(5) $(x+2y)y' = 1$.　$y(0) = -1$

(6) $x^2y' - \cos 2y = 1$　$y(+\infty) = \frac{9\pi}{4}$

(7) $3y^2y' + 16x = 2xy^3$.　当 $x \to +\infty$ 时, $y(x)$ 有界.

(1) ① $\frac{dy}{y} = -\frac{x}{x+1}dx = (-1+\frac{1}{x+1})dx \Rightarrow \ln|y| = -x + \ln|x+1| + C_0 \Rightarrow y = C\frac{x+1}{e^x}$ $(C \neq 0)$

② $y = 0$　综上: $y = \frac{x+1}{e^x}$

(2) ① $-\frac{dy}{y^2} = \frac{2x}{x^2-1} \Rightarrow \frac{1}{y} = \ln|x^2-1| + C \Rightarrow y = \frac{1}{C+\ln|x^2-1|} \Rightarrow y = \frac{1}{\ln|x^2-1|}$

(3). $x\frac{dy}{dx} = y^2 - y \Rightarrow \frac{dy}{y^2-y} = \frac{1}{x} \Rightarrow \ln\left|\frac{y-1}{y}\right| = \ln|x| + C \Rightarrow y = \frac{1}{1+x}$

(4). 令 $u = 4x+2y-1$　则 $u' = 4+2\sqrt{u} \Rightarrow \frac{du}{4+2\sqrt{u}} = dx \Rightarrow \sqrt{u} - 2\ln(2+\sqrt{u}) = x + C$.

$x=0$ 时, $u=1$. $\sqrt{4x+2y-1} - 2\ln\left(2+\sqrt{4x+2y-1}\right) = x + 1 - 2\ln 3$.

(5) 令 $u = x+2y$　则 $u' = 1+2y' = 1+\frac{2}{u}$. $\Rightarrow \frac{u\,du}{u+2} = dx \Rightarrow u - 2\ln|u+2| = x + C$

或 $u = -2$. 结合初值得到 $x+2y+2 = 0$

(6). $y' = (\cos 2y + 1)\frac{1}{x^2} \Rightarrow \frac{dy}{\cos 2y+1} = \frac{dx}{x^2}$　$x \to +\infty$. $\cos 2y + 1 \to \cos\frac{\pi}{2} + 1 \neq 0$.

$\Rightarrow \frac{dy}{2\cos^2 y} = \frac{dx}{x^2}$. $\Rightarrow \frac{1}{2}\tan y = -\frac{1}{x} + C \Rightarrow y = \arctan\left(\frac{x}{x-2}\right) + 2\pi$

(7) ① $y' = \frac{2xy^3 - 16x}{3y^2}$　令 $u = \frac{y}{x}$ $\Rightarrow y' = u'x + u = \frac{2u^3 - 16}{3u^2} \Rightarrow u' = -\frac{u^3+16}{3u^2}\frac{1}{x}$

1°. $y \neq -x$ $\Rightarrow \frac{3u^2\,du}{u^3+16} = -\frac{dx}{x} \Rightarrow \ln|u^3+16| = -\ln(C|x|) \Rightarrow (\frac{y}{x})^3 + 16 = C\frac{1}{x}$

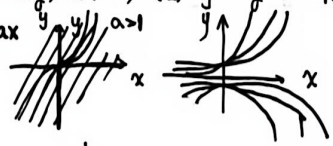
$\Rightarrow y = \left(\sqrt[3]{\frac{C}{x} - 16}\right)x$

2°. $y = x$.

② $y \equiv 0 \Rightarrow y \equiv 0$

Ⅱ.1.2. 解方程, 并作出积分曲线. (1) $y' = ay$, $a \neq 0$; (2) $y' = y^a$. 其 $a = \frac{1}{5}, 1, 2$

(1) $\ln|y| = ax + C_0 \Rightarrow y = Ce^{ax}$



(2) $\frac{1}{1-a}y^{-a+1} = x + C \Rightarrow y = (1-a)^{\frac{1}{1-a}}(x+C)^{\frac{1}{1-a}}$

Ⅱ.1.3. $y' = f(y)$ 其中 $f(y)$ 在 $y=a$ 的某邻域内连续, 且 $f(y) = 0 \Leftrightarrow y = a$.

证明: 对于 $y=a$ 上任一点 (x_0, a), 方程满足初值 $y(x_0) = a$ 的解存在且唯一的充要条件为 $\left|\int_a^{a+\varepsilon}\frac{1}{f(y)}dy\right| = +\infty$

$\forall\, \varepsilon > 0$.

$\Rightarrow$:

证明: $y = y(x)$, 满足 $y(x_0) = a$, $f(y) = 0$

注意到 $y \equiv a$ 为解. 若 $\exists\, \varepsilon > 0$. s.t. $\left|\int_a^{a+\varepsilon}\frac{1}{f(y)}dy\right| < +\infty$.

则 对 $\frac{dy}{f(y)} = 1$ 两边从 a 到 y 积分. $a < y < a+\varepsilon$　$\int_a^y \frac{dt}{f(t)} = x - a$ 右侧单调, 由反函数定理

$\exists\ y=y(x)\neq a$. s.t. $y(x)$ 满足方程 矛盾!

反之: 若 $\forall \varepsilon>0 \quad \left|\int_0^{a+\varepsilon}\frac{1}{f(y)}dy\right|=+\infty$. 若有 $y(x)\neq a$ 则 $\left|\int_{y(x)}^{a+\varepsilon}\frac{y'(x)\,dx}{f(y(x))}\right|=\left|y^{-1}(a+\varepsilon)-y^{-1}(a)\right|<+\infty$

取 $a+\varepsilon\in \operatorname{Im} y(x)$

矛盾! □

Ⅱ.1.4. $f(x,y)$ 称为 d 次拟齐次函数. 若 $f(t^{\alpha s}x, t^{\beta s}y)=t^{ds}f(x,y)$, $t>0$, $\alpha+\beta=1$ 为正常数 称为权 ($\forall s\in\mathbb{R}$, 有)

考虑 $P(x,y)\,dx+Q(x,y)\,dy=0$. $P(x,y)$ 为 d_0 次拟齐次函数, $Q(x,y)$ 为 d_1 次拟齐次函数.

x 与 y 的权为 α,β. 证明: 当 $d_0-d_1=\beta-\alpha$ 时, 方程可用初等积分求解.

证明: 令 $y=x^{\frac{\beta}{\alpha}}u$. 则 $x^{\frac{d_0}{\alpha}}P(1,u)\,dx+x^{\frac{d_1}{\alpha}}Q(1,u)\left(x^{\frac{\beta}{\alpha}}u'+u\cdot\frac{\beta}{\alpha}x^{\frac{\beta}{\alpha}-1}\right)=0$

注意到 $x^{\frac{d_0}{\alpha}}=x^{\frac{d_1}{\alpha}+\frac{\beta}{\alpha}-1}$ 故 $x^{\frac{d_0}{\alpha}}\left(P(1,u)+\frac{\beta u}{\alpha}\right)dx=-x^{\frac{d_1+\beta}{\alpha}}Q(1,u)\,u'$

当 $P(1,u)+\frac{\beta}{\alpha}u\neq 0 \Rightarrow \frac{Q(1,u)\,du}{P(1,u)+\frac{\beta}{\alpha}u}=\frac{-dx}{x}$ □

Ⅱ.1.5. 证明: $y'=\sqrt[3]{\frac{y^2+1}{x^4+1}}$ 的每条积分曲线都有2条水平渐近线.

证明: $y'>0 \Rightarrow y=y(x)\uparrow$. $F(y)=\int_{y_0}^{y(x)}\frac{dy}{(y^2+1)^{\frac{1}{3}}}=\int_{x_0}^{x}\frac{dt}{(t^4+1)^{\frac{1}{3}}}$

由p-积分判别法, $\lim_{x\to+\infty}\int_{x_0}^{x}\frac{dt}{(t^2+1)^{\frac{1}{3}}}=C_1<\infty$

$\lim_{x\to-\infty}\int_{x}^{x_0}\frac{dt}{(t^2+1)^{\frac{1}{3}}}=C_2<\infty$

$\Rightarrow \lim_{x\to+\infty}y(x)=\lim_{x\to+\infty}F^{-1}(F(y(x)))=F^{-1}(C_1)$

$\lim_{x\to-\infty}y(x)=\lim_{x\to-\infty}F^{-1}(F(y(x)))=F^{-1}(C_2)$ □.

Ⅱ.1.6. $f(x)$ 在区间 $[1,+\infty)$ 上连续且恒正, 证明: 若 $\int_1^x f(t)\,dt\leq(f(x))^3$

则 $f(x)\geq\sqrt{\frac{2}{3}(x-1)}$

证明: 令 $\varphi(x)=\int_1^x f(t)\,dt$. 则 $\varphi(x)\leq(\varphi'(x))^3 \Rightarrow dx\leq\varphi^{-\frac{1}{3}}d\varphi$, $\varphi(1)=0$

$\Rightarrow \frac{3}{2}\varphi^{\frac{2}{3}}\geq x-1 \Rightarrow \varphi(x)\geq\left[\frac{2}{3}(x-1)\right]^{\frac{3}{2}}$

$\Rightarrow \varphi'(x)=f(x)\geq\frac{3}{2}\left[\frac{2}{3}(x-1)\right]^{\frac{1}{2}}\geq\sqrt{\frac{2}{3}(x-1)}$ □

Ⅱ.1.7. 解方程: (1) $xy'-2y=2x^4$; (2) $(2x+1)y'=4x+2y$

(3) $(xy+2e^x)dx-x\,dy=0$; (4) $2x(x^2+y)dx=dy$

(5) $(1-2xy)y'=y(y-1)$

解: (1) ① $y'-2\frac{y}{x}=2x^3$

两边同乘 $\frac{1}{x^2}$: $\frac{1}{x^2}\frac{dy}{dx}-2\frac{y}{x^3}=2x$

$\Rightarrow \frac{d}{dx}\left(\frac{1}{x^2}y\right)=-\frac{2}{x^3}y+\frac{1}{x^2}\frac{dy}{dx}=2x$

$\frac{y}{x^2}=x^2+C \Rightarrow y=x^4+Cx^2$

②

(2) $\frac{dy}{dx}-\frac{2}{2x+1}y=\frac{4x}{2x+1}$ 两边同乘 $\frac{1}{2x+1}$ $(x>-\frac{1}{2})$

$\Rightarrow \frac{d}{dx}\left(\frac{1}{2x+1}y\right) = -\frac{2}{(2x+1)^2}y + \frac{dy}{dx}\cdot\frac{1}{2x+1} = \frac{4x}{(2x+1)^2}$ $\int \frac{4x\,dx}{(2x+1)^2} = \int \frac{2(t-1)\,dt}{t^2} = \ln t + \frac{1}{t}$

$\Rightarrow \frac{y}{2x+1} = $ $= \ln(2x+1) + \frac{1}{2x+1}$.

$\Rightarrow y = (2x+1)\ln(2x+1) + 1$

同理可得 $x<-\frac{1}{2}$ 时的值 $x=-\frac{1}{2}$ 时定义 $y=1$.

(3) $\frac{dy}{dx} - y = \frac{e^x}{x}$ 两边同乘 e^{-x} 得 $\frac{d}{dx}(e^{-x}y) = e^{-x}\frac{dy}{dx} - e^{-x}y = \frac{1}{x}$

$\Rightarrow \frac{y}{e^x} = \ln|x| + C. \Rightarrow y = (C+\ln|x|)e^x$

(4). $\frac{dy}{dx} - 2xy = 2x^3$ 两边同乘 e^{-x^2} 得 $\frac{d}{dx}(e^{-x^2}y) = e^{-x^2}(-2xy) + e^{-x^2}\frac{dy}{dx} = 2x^3e^{-x^2}$

$\Rightarrow e^{-x^2}y = \int 2x^3e^{-x^2}dx = -\int x^2\,d(e^{-x^2}) = -x^2e^{-x^2} + \int 2xe^{-x^2}dx = -x^2e^{-x^2} - e^{-x^2} + C$

$\Rightarrow y = Ce^{x^2} - x^2 - 1$

(5). $(1-2xy)y' = y(y-1)$ $\Rightarrow \frac{dx}{dy} + \frac{2}{y-1}x = \frac{1}{y(y-1)}$.

1°. $y\neq 0$ 或 1.

两边同乘 $(y-1)^2$ $\frac{d}{dy}(x(y-1)^2) = \frac{dx}{dy}(y-1)^2 + 2(y-1)x$

$= \frac{y-1}{y} = 1 - \frac{1}{y}$

$\Rightarrow x(y-1)^2 = y - \ln|y| + C$

2°. $y\equiv 0$ 或 1 □.

2. 求微分方程 $y'\sin 2x = 2(y+\cos x)$ 在当 $x\to\frac{\pi}{2}$ 时有界的解

解: $y' - \frac{2}{\sin 2x}y = \frac{1}{\sin x}$ 两边同乘 $\tan\frac{x}{2}\cos x$ 得 $\frac{d}{dx}\left(\tan\frac{x}{2}\cos x\, y\right) = \frac{\tan\frac{x}{2}}{\tan x}$

$\Rightarrow \tan\frac{x}{2}\cos x\, y = $

$\int \frac{\tan\frac{x}{2}}{\tan x}dx = x - \tan\frac{x}{2} + C$. □

3. 设 $xy' + ay = f(x)$, $a>0$. $\lim_{x\to 0} f(x) = b$. 证明: 当 $x\to 0$ 时, 该方程仅一个有界的解. 并求解在 $x\to 0$ 时极限

证明: $y' + \frac{a}{x}y = \frac{f(x)}{x}$ 两边同乘 x^a. $\Rightarrow \frac{d}{dx}(x^a y) = x^{a-1}f(x)$ 设 $F(x) = \int_0^x t^{a-1}f(t)\,dt$

$\Rightarrow x^a y = \int x^{a-1}f(x)dx \Rightarrow y = \frac{1}{x^a}\int x^{a-1}f(x)\,dx$ 则 $F(0) = 0$

$= \frac{1}{x^a}(F(x) + C)$.

当 $x\to 0$ 时 y 有界 $\Leftrightarrow C = 0$ 故 $y(x) = \frac{1}{x^a}\int_0^x t^{a-1}f(t)\,dt$ $\lim_{x\to 0} y(x) = \frac{b}{a}$ (洛必达) □.

4. 求 $y' = 2y\cos^2 x - \sin x$ 的周期解.

解: $y' - 2y\cos^2 x = -\sin x$ 两边同乘 $e^{-x-\frac{\sin 2x}{2}}$

$\Rightarrow \frac{d}{dx}\left(\frac{y}{e^{x+\sin x\cos x}}\right) = \frac{-\sin x}{e^{x+\sin x\cos x}}$

5. $f(t)$ 连续，$|f(t)| \le M$.$(\forall t \in \mathbb{R})$ 证明：$\frac{dx}{dt} + x = f(t)$ 在 $t \in \mathbb{R}$ 上只有一个有界解；若 $f(t)$ 周期，则有界解也是周期的.

证明：两边同乘 e^t. $\Rightarrow \frac{d}{dt}(e^t x) = f(t)e^t \quad \Rightarrow e^t x = \int_0^t f(v)e^v dv + C$

$$x = e^{-t}\left(\int_0^t f(v)e^v dv + C\right)$$